

Kumar Dhananjaya Shivanna

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EDUCATION

University of Sydney

Master of Computer Science (Advanced Entry)

Sydney, Australia
Feb 2026 – Nov 2027 (Expected)

Maharaja Institute of Technology Thandavapura

Bachelor of Engineering in Computer Science (CGPA: 8.60/10)

Mysuru, India
2019 – 2023

TECHNICAL SKILLS

Languages: Go, C++, Java, Python, TypeScript, JavaScript (ES6+), SQL (PostgreSQL, MySQL)
Systems & Networking: Node.js, Redis, gRPC, WebSockets, TCP/IP, Unix/Linux, Microservices, Zero-Trust
AI & Machine Learning: Agentic AI, LLM Integrations, Prompt Engineering, Model Context Protocol (MCP)
Cloud & DevOps: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, Helm, CI/CD (GitHub Actions)
Frontend: React, Next.js (SSR/ISR), Android Development, Redux Toolkit, Tailwind CSS

WORK EXPERIENCE

Examic EdTech Pvt. Ltd.

Associate Software Engineer

Mysuru, India
Jul 2024 – Feb 2026

- Sustained sub-150ms bidirectional telemetry latency for **5,000+** concurrent sessions by architecting a real-time assessment engine on **Azure** utilizing **WebSockets** and event-driven data propagation.
- Reduced manual operational overhead by **40%** by engineering an enterprise CMS implementing **Optimistic Locking** and dynamic schema validation to automate large-scale educational data ingestion.
- Decreased p95 API response times by **25%** by optimizing **MongoDB aggregation pipelines** and compound indexing strategies, facilitating high-availability for mission-critical examination traffic.

Kandra Digital Pvt. Ltd.

MERN-Stack Developer Intern

Bengaluru, India
Feb 2024 – Jul 2024

- Accelerated First Contentful Paint (FCP) by **30%** and improved organic SEO visibility by developing production-grade SSR applications utilizing **Next.js** and optimized asset delivery pipelines.
- Eliminated configuration drifts across CI/CD deployment pipelines by containerizing application services using **Docker** to enforce strict development and production environment parity.

OPEN SOURCE CONTRIBUTIONS

Komodor (komodorio) | Go, Kubernetes, Helm, Cloud-Native

Jan 2026

- Engineered CLI configuration flags to bypass slow health checks (PR #493), significantly reducing CI/CD execution latency for deployments in high-latency environments.
- Patched critical Helm chart deployment bugs by resolving incorrect upgrade recommendations (PR #577) and preventing redundant rollback mutations (PR #569).

PROJECTS

PulseTrace | Node.js, Redis, BullMQ, PostgreSQL, React, TypeScript

- Architected a decoupled telemetry pipeline leveraging Node.js and BullMQ sidecar workers, reliably processing **1M+** asynchronous events with minimal latency to power a 24-hour continuous uptime audit trail.
- Engineered a high-performance React dashboard and custom SDK, reducing client integration to under **60 seconds** while delivering real-time observability into Core Web Vitals (LCP, INP, CLS).

Zero-Trust API Platform | NestJS, PostgreSQL, Redis, JWT, Docker, AWS EC2

- Enforced strict Zero-Trust isolation for **100%** of ingress traffic by architecting a centralized NestJS Gateway and 5 independent microservices to handle cryptographic JWT validation.
- Achieved decoupled access control and 100% authorization visibility by engineering a Policy Decision Point (PDP) with **Policy-as-Code** for RBAC/ABAC execution, feeding a real-time React UI.

NexusEdit | React, Node.js, Socket.io, Redis, CRDT (Yjs), MongoDB

- Engineered a real-time collaborative engine ensuring **100%** conflict-free document mutations using **Conflict-free Replicated Data Types (CRDTs)** for sub-second live cursor tracking.
- Architected a horizontally scalable synchronization pipeline maintaining **<50 ms** cross-server messaging latency via **Redis Pub/Sub**, paired with asynchronous MongoDB document snapshotting.

LEADERSHIP & ACHIEVEMENTS

- Directed **ACES & Clusteroids** engineering clubs as President (2022-23), spearheading technical initiatives and acting as Lead Organizer for a 24-hour Inter-college Hackathon supporting **100+** participants.
- Demonstrated technical excellence by securing **1st Place** at the Inter-College Web Dev Competition, **2nd Place** at the HPE SWARM-IT National Hackathon, and the **Best Idea Award** at the GND National Technical Symposium.